



NORTH SYDNEY BOYS HIGH SCHOOL

2010 YEAR 7 COMMON TEST 2

Mathematics

Student Name: _____

Class: _____

Teacher _____

General Instructions

- Working time – 50 minutes
- Write your answers in the space provided.
- Write using blue or black pen.
- No calculators are allowed.
- All necessary working should be shown.
- Marks may not be awarded for carelessly or badly arranged work.

(To be used by the exam markers only.)

Question	1 Number	2 Angles	3 W.M	Total	Total %
Mark	<u>35</u>	<u>25</u>	<u>11</u>	<u>71</u>	<u>100</u>

SECTION 1

Question 1

- a) Express 7% as a decimal:(1)
- b) Express 0.001 as a percentage:(1)
- c) Express 85% as a fraction in simplest form:

(2)

Question 2

- a) Round 19749.27645 to:
- i) the nearest thousand:.....(1)
- ii) the nearest hundredth:.....(1)
- b) Write the following numbers in **ascending** order:
 53% , 0.06 , $\frac{7}{2}$, 3 , 200% , 0.010 , 0.7%
(2)

Question 3

Write $5 + \frac{21}{1000}$ as a decimal:
(1)

Question 4

Evaluate each of the following (show **all** working):

- | | |
|--|---|
| a) $17.01 + 3.782$
.....
.....
.....(1) | b) $191.2 - 1.83$
.....
.....
.....(1) |
|--|---|

c) 12.1×7

.....

(1)

d) 7.16×0.92

.....

(2)

e) $82.2 \div 6$

.....

(1)

f) $0.0072 \div 0.012$

.....

(2)

Question 5

Simplify the following fractions down to their lowest terms. Express your answer as a mixed numeral where applicable.

a) $\frac{6}{15}$

.....

(1)

b) $\frac{72}{48}$

.....

(2)

Question 6

What percentage is \$8 of \$64?.....

.....

(2)

Question 7

Evaluate each of the following. Simplify your answer as much as possible:

a) $\frac{2}{6} + \frac{1}{6}$

.....

(2)

b) $\frac{7}{8} - \frac{1}{3}$

.....

(2)

c) $3\frac{2}{3} - 2\frac{3}{4}$

.....

(3)

d) $\frac{2}{7} \div 1\frac{1}{5}$

.....

(2)

e) 30% of \$25

.....

(2)

f) $6 \div 11$ (express your answer as a decimal)

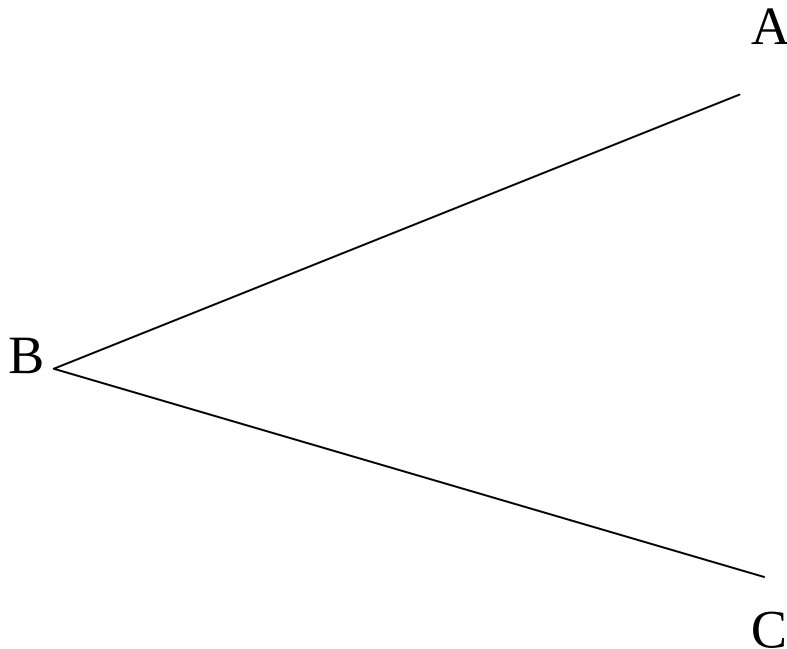
.....

(2)

End of Section 1 – Proceed to Section 2

SECTION 2

Question 1



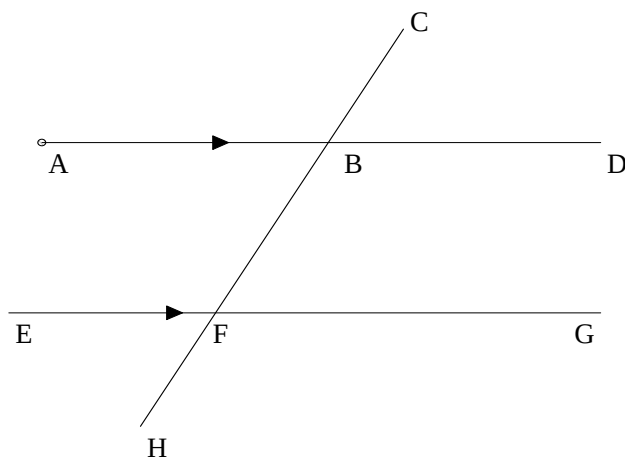
- a) Name the angle in the above diagram:.....(1)
- b) What type of angle is this?.....(1)
- c) Name the two intervals in the above diagram:.....(2)
- d) Use your compass and a straight-edge to bisect this angle (show all construction lines). (2)

Question 2

Express the following as True bearings.

- a) N 30° E =(1)
- b) S 10° W =(1)
- c) Due West =(1)

Question 3



From the diagram above, name a pair of :

- a) Corresponding angles:.....(1)
- b) Alternate angles:.....(1)
- c) Co-interior angles:.....(1)
- d) Vertically opposite angles:.....(1)
- e) Choose the correct word from the list below to complete the following sentence:

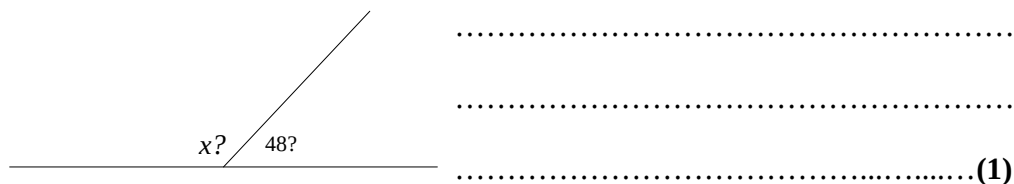
$\angle ABC$ is to $\angle ABF$ (1)

alternate , equal , co-interior , supplementary , corresponding , complimentary

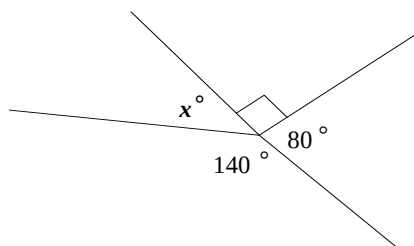
Question 4

Calculate the value of x° in each of the following diagrams (*not to scale*).

a)



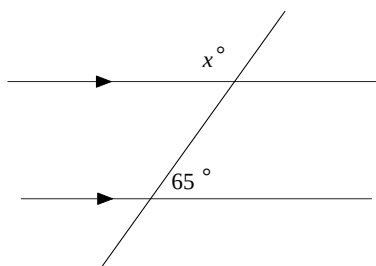
b)



.....

(2)

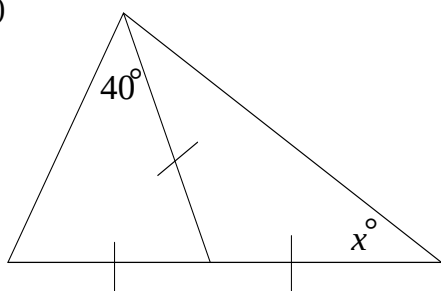
c)



.....

(1)

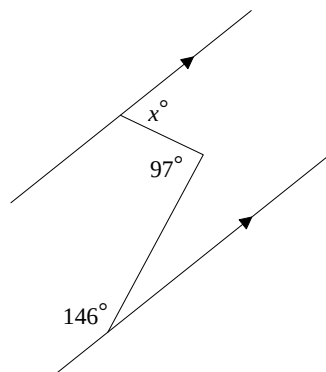
d)



.....

(3)

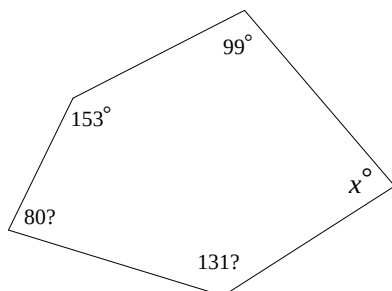
e)



.....

(2)

f)



.....

(2)

End of Section 2 – Proceed to Section 3

SECTION 3

Question 1

In the space below, use your compass and ruler to construct a triangle with side lengths 5 cm, 6 cm and 7cm. (2)

Measure the size of the largest angle of this triangle.....(1)

Question 2

A new jacket cost \$90.20 after a 10% tax was added. How much would the jacket cost without the tax?

.....
.....
.....
.....
.....(2)

Question 3

A computer costs \$800. It is then marked up by 20%. If I get a 10% discount when I buy the computer how much do I pay?

.....

.....

.....

.....

.....(2)

Question 4

After Simon's birthday, three fifths of the cake has been eaten. The next day Simon eats one sixth of the remaining cake:

a) What fraction of the cake is now left?

.....

.....

.....

.....

.....(2)

b) Overall 800 grams of the cake has been eaten. How much did the cake originally weigh?

.....

.....

.....

.....(2)

End of Examination

Suggested Solutions**Section 1 Fractions, Decimals & Percentages****Question 1**

(a) (1 mark)

$$7\% = 0.07$$

(c) (1 mark)

(b) (1 mark)

$$0.001 = 0.1\%$$

(c) (2 marks)

$$\frac{85}{100} \div \frac{5}{5} = \frac{17}{20}$$

(d) (2 marks)

$$\begin{array}{r} 17 \\ 5 \overline{)85} \\ \underline{50} \\ 35 \\ \underline{35} \\ 0 \end{array}$$

(e) (1 mark)

Question 2

(a) i. (1 mark)

$$19\,749.276\,45 = 20\,000 \text{ (nearest 1 000)}$$

ii. (1 mark)

$$19\,749.276\,45 = 19\,749.28 \text{ (nearest 100th)}$$

(b) (2 marks)

✓ [-1] for each mistake.

(f) (2 marks)

$$0.7\% \quad 0.010 \quad 0.06 \quad 0.53 \quad 200\% \quad 3 \frac{7}{2}$$

$$0.007\,2 \div 0.012 = 7.2 \div 12 = 0.6$$

Question 3 (1 mark)

$$5 + \frac{21}{1000} = 5.021$$

Question 4

(a) (1 mark)

$$\begin{array}{r} 1 \\ 1\,7.0\,1 \\ + \quad 3.7\,8\,2 \\ \hline 2\,0.7\,9\,2 \end{array}$$

Question 5

(a) (1 mark)

$$\begin{array}{r} 1\,9\,1.2\,0 \\ - \quad 1.8\,3 \\ \hline 1\,8\,9.3\,7 \end{array}$$

$$\begin{array}{r} 1\,2.1 \\ \times \quad 7 \\ \hline 8\,4.7 \end{array}$$

$$\begin{array}{r} 7.1\,6 \\ \times 0.9\,2 \\ \hline 1\,4\,3\,2 \\ 6\,4\,4\,4 \\ \hline 6.5\,8\,7\,2 \end{array}$$

$$\begin{array}{r} 137 \\ 6 \overline{)822} \\ \underline{600} \\ 222 \\ \underline{180} \\ 42 \\ \underline{42} \\ 0 \end{array}$$

$$\begin{array}{r} 6 \\ 12 \overline{)72} \\ \underline{72} \\ 0 \end{array}$$

$$\frac{6}{15} \div \frac{3}{5} = \frac{2}{5}$$

(b) (2 marks)

$$\frac{72 \div 24}{48 \div 24} = \frac{3}{2}$$

Question 6 (2 marks)

$$\frac{\$8}{\$64} = \frac{1}{8} = 0.125$$

Hence \$8 is 12.5% of \$64.

Question 7

(a) (2 marks)

$$\begin{aligned} \frac{2}{6} + \frac{1}{6} &= \frac{3}{6} \\ &= \frac{1}{2} \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} \frac{7 \times 3}{8 \times 3} - \frac{1 \times 8}{3 \times 8} &= \frac{21}{24} - \frac{8}{24} \\ &= \frac{13}{24} \end{aligned}$$

(c) (3 marks)

$$\begin{aligned} 3\frac{2}{3} - 2\frac{3}{4} &= \frac{11}{3} \times \frac{4}{4} - \frac{11}{4} \times \frac{3}{3} \\ &= \frac{44}{12} - \frac{33}{12} \\ &= \frac{11}{12} \end{aligned}$$

(d) (2 marks)

$$\begin{aligned} \frac{2}{7} \div 1\frac{1}{5} &= \frac{2}{7} \div \frac{6}{5} \\ &= \frac{2}{7} \times \frac{5}{6} \\ &= \frac{5}{21} \end{aligned}$$

(e) (2 marks)

$$\begin{array}{r} 25 \\ \times 0.3 \\ \hline 75 \\ 075 \\ \hline \end{array}$$

(f) (2 marks)

$$6 \div 11 = \frac{11}{6} = 0.\overline{54}$$

$$\begin{array}{r} 5454 \\ 11 \overline{) 60000} \\ \underline{55000} \\ 5000 \\ \underline{4400} \\ 600 \\ \underline{550} \\ 50 \\ \underline{44} \\ 6 \end{array}$$

Section 2 Angles**Question 1**

(a) (1 mark)

 $\angle ABC$

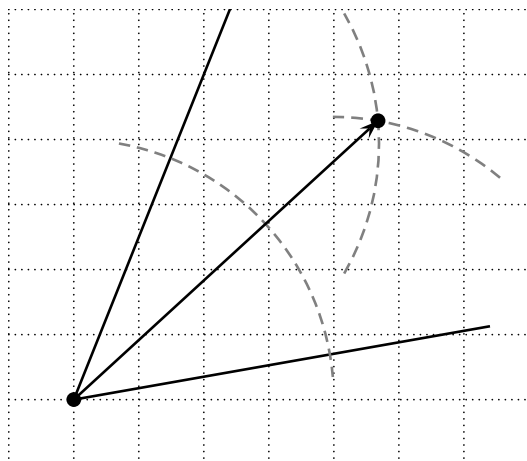
(b) (1 mark) – acute

(c) (2 marks)

 AB, BC

(d) (2 marks) – General rule to bisect an angle:

- 1.) With the compass needle placed at the vertex, draw an arc that intersects with the two arms that subtend the angle.
- 2.) With the compasses open, place the needle at the intersection of the arc & one of the arms. Draw an arc from that point.
- 3.) With the compasses open at the same length, place the needle at the intersection of the arc & the other arm. Draw an arc from that point.
- 4.) Join the intersection of the two arcs & the vertex of the angle to be bisected.

**Question 2**

(a) (1 mark)

$$\text{N}30^\circ\text{E} = 030^\circ\text{T}$$

(b) (1 mark)

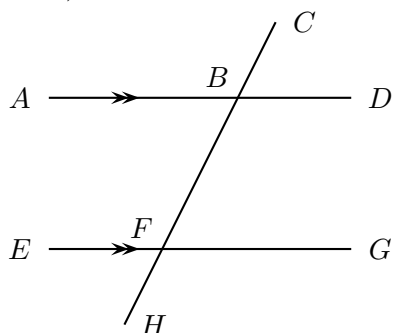
$$\text{S}10^\circ\text{W} = 190^\circ\text{T}$$

(c) (1 mark)

$$\text{Due west} = 270^\circ\text{T}$$

Question 3

(a) (1 mark)



These are the permitted pairs of *corresponding* $\angle$. Only one pair will be sufficient.

- $\angle ABC, \angle EFB$. • $\angle ABF, \angle EFH$.
- $\angle DBC, \angle GFB$. • $\angle DBF, \angle GFH$.

(b) (1 mark)

These are the permitted pairs of *alternate* $\angle$. Only one pair will be sufficient.

- $\angle ABF, \angle BFG$. • $\angle DBF, \angle BFE$.

(c) (1 mark) These are the permitted pairs of *cointerior* $\angle$. Only one pair will be sufficient.

- $\angle ABF, \angle BFE$. • $\angle DBF, \angle BFG$.

(d) (1 mark) These are the permitted pairs of *vertically opposite* $\angle$. Only one pair will be sufficient.

- $\angle ABC, \angle FBD$. • $\angle EFB, \angle GFH$.
- $\angle ABF, \angle CBD$. • $\angle EFH, \angle BFG$.

(e) (1 mark)

$\angle ABC$ is **supplementary** to $\angle ABF$.

Question 4

(a) (1 mark)

$$\begin{aligned} x &= 180^\circ - 48^\circ \\ &= 132^\circ \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} x &= 360^\circ - 90^\circ - 140^\circ - 80^\circ \\ &= 270^\circ - 220^\circ \\ &= 50^\circ \end{aligned}$$

(c) (1 mark)

$$\begin{aligned} x &= 180^\circ - 65^\circ \\ &= 115^\circ \end{aligned}$$

(d) (3 marks)

$$\begin{aligned} x &= \frac{180 - (2 \times 40^\circ)}{2} \\ &= 50^\circ \end{aligned}$$

(e) (2 marks)

$$\begin{aligned} x &= 97^\circ - (180^\circ - 146^\circ) \\ &= 63^\circ \end{aligned}$$

(f) (2 marks)

$$\begin{aligned} x &= 540^\circ - 99^\circ - 131^\circ - 153^\circ - 80^\circ \\ &= 540^\circ - 230^\circ - 153^\circ - 80^\circ \\ &= 540^\circ - 310^\circ - 153^\circ \\ &= 77^\circ \end{aligned}$$

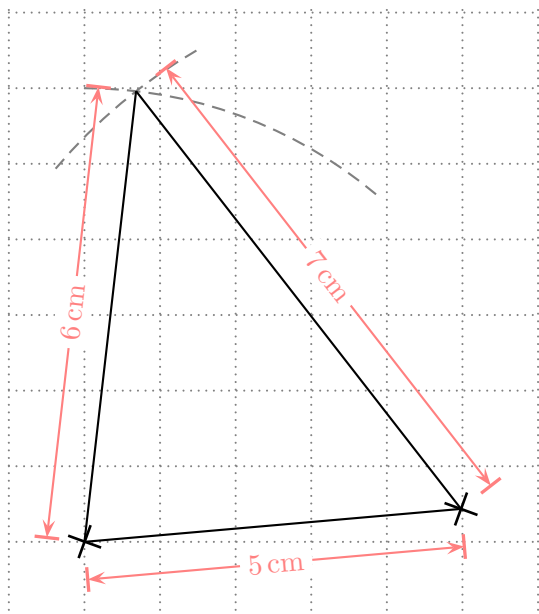
Section 3 Working Mathematically

Question 3 (2 marks)

Question 1

$$\$800 \times 1.2 \times 0.9 = 800 \times 1.08$$

(a) (2 marks)



$$\begin{array}{r} 1.2 \\ \times 0.9 \\ \hline 108 \\ 1.08 \end{array} \quad \begin{array}{r} 800 \\ \times 1.08 \\ \hline 6400 \\ 8000 \\ \hline 864.00 \end{array}$$

Hence the new cost is \$864.

Question 4

(a) (2 marks)

- After Simon's birthday, three fifths of the cake has been eaten.
 $\Rightarrow \frac{2}{5}$ remaining.
- The next day Simon eats one sixth of the *remaining* cake.
 $\Rightarrow \frac{5}{6}$ of the uneaten cake remains.

(b) (1 mark)

$$78^\circ \pm 2^\circ$$

(using the cosine rule will result in $\approx 78^\circ$)

Question 2 (2 marks)

$$\begin{aligned} 1.1x &= \$90.2 \\ x &= \frac{90.2}{1.1} = \frac{902}{11} \end{aligned}$$

$$\begin{array}{r} 82 \\ 11 \overline{) 902} \\ \underline{880} \\ 22 \\ \underline{22} \\ 0 \end{array}$$

$\therefore x = \$82$

$$\frac{2}{5} \times \frac{5}{6} = \frac{2}{6} = \frac{1}{3}$$

 $\frac{1}{3}$ of the entire cake remains.

(b) (2 marks)

$$\begin{aligned} \frac{2}{3}x &= 800 \text{ g} \\ x &= \frac{800 \times 3}{2} \\ &= 400 \times 3 = 1.2 \text{ kg} \end{aligned}$$